Linear Algebra Problem Sheet-2

On Dual Spaces, Quotient Spaces, and Direct Sums

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Abstract

The first problem sheet for Graduate Mathematical Toolkit-1 2026. Please contact the TAs for doubts at sunain.ug2023@cmi.ac.in and Avishekh@cmi.ac.in.

Problem 0.1. *You have probably already seen this in class but it is always a good idea to go over it again. Let V be an n-dimensional vector space with basis $\mathcal{B} = (v_1, \ldots, v_n)$.*

1. *Prove that there exists a unique basis $\mathcal{B}^* = (v^1, \ldots, v^n)$ of V^* satisfying*

$$v^i(v_j) = \delta_{ij}.$$

where $\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$

2. *If $\varphi = a_1 v^1 + \cdots + a_n v^n \in V^*$, determine a_i in terms of φ and v_i.*

3. *Let $T : V \to W$ be linear. The dual operator $T^* : W^* \to V^*$ is defined so that for any $w^* \in W^*$ and $v \in V$ we have,*

$$T^*(w^*)(v) = w^*(T(v))$$

Suppose you are given the matrix of T in some basis of V and W.Describe the matrix of $T^* : W^* \to V^*$ with respect to the dual bases of chosen bases of V and W.

Problem 0.2. *Let $U \leq V$. The* annihilator *of U is*

$$U^\circ = \{\varphi \in V^* : \varphi(u) = 0 \text{ for every } u \in U\}.$$

1. *Prove that U° is a subspace of V^*.*

2. *Prove that* $\dim U^\circ = \dim V - \dim U$.

3. *Suppose $U = \operatorname{span}(u_1, \ldots, u_r)$, where the u_i are linearly independent. Show that*

$$U^\circ = \{\varphi \in V^* : \varphi(u_1) = \cdots = \varphi(u_r) = 0\}.$$

4. *If $U \leq W \leq V$, prove that $W^\circ \subseteq U^\circ$.*

Problem 0.3. *Let $T : V \to W$ be linear.*

1. *Prove that* $\ker T^* = (imT)^\circ$.

2. *Prove that* $im T^* = (\ker T)^\circ$.
3. *Using the above and the previous problem show that* $\operatorname{rank}(T^*) = \operatorname{rank}(T)$.
4. *Show that* T *is injective if and only if* T^* *is surjective, and that* T *is surjective if and only if* T^* *is injective.*

Problem 0.4. *Let* $U \leq V$*, and let* $\pi : V \to V/U$*,* $v \mapsto v + U$*, be the canonical projection.*

1. *Prove that* π *is linear and surjective.*
2. *Prove that* $\ker \pi = U$.
3. *Let* $T : V \to W$ *be linear and suppose* $U \subseteq \ker T$*. Prove that there exists a unique linear map* $\widetilde{T} : V/U \to W$ *such that* $T = \widetilde{T} \circ \pi$.

Problem 0.5. *Let* $U \leq V$.

1. *Prove the dimension formula* $\dim(V/U) = \dim V - \dim U$.
2. *Let* $W \leq V$*. Prove that* $(U + W)/U \cong W/(U \cap W)$.
3. *Deduce the dimension formula*

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

Problem 0.6. *Let* $U \leq V$.

1. *Show that every* $\psi \in (V/U)^*$ *determines a functional* $\psi \circ \pi \in V^*$ *that vanishes on* U.
2. *Prove that the map* $\Phi : (V/U)^* \to V^*$*,* $\psi \mapsto \psi \circ \pi$*, is injective.*
3. *Prove that* $im \Phi = U^\circ$.
4. *Conclude that there is a canonical isomorphism* $(V/U)^* \cong U^\circ$.

Problem 0.7. *Some computation : Let* $V = \mathbb{R}^4$ *and* $U = \operatorname{span}\{(1,0,1,0),(0,1,0,1)\}$.

1. *Find a basis for* V/U.
2. *Define a linear functional* $\varphi(x_1, x_2, x_3, x_4) = a_1x_1 + a_2x_2 + a_3x_3 + a_4x_4$*. Determine the conditions on the* a_i *for* φ *to belong to* U°.
3. *Find a basis for* U°.

Problem 0.8. *Let* $U, W \leq V$*. Prove that the following are equivalent:*

1. *Every* $v \in V$ *can be written uniquely as* $v = u + w$*,* $u \in U$*,* $w \in W$.
2. $V = U + W$ *and* $U \cap W = \{0\}$.
3. *The map* $U \oplus W \to V$*,* $(u, w) \mapsto u + w$*, is an isomorphism.*
4. *Prove that* $\dim V = \dim U + \dim W$.

Problem 0.9. *Suppose $V = U \oplus W$. Define $P : V \to V$ by writing $v = u + w$ uniquely and setting $P(v) = u$.*

1. *Prove that P is linear.*
2. *Prove that $P^2 = P$.*
3. *Show that $imP = U$ and $\ker P = W$.*
4. *Define $Q = I - P$. Show that Q is also a projection and determine its image and kernel.*
5. *Prove that every projection $P : V \to V$ gives rise to a direct-sum decomposition $V = imP \oplus \ker P$.*

Problem 0.10. *Suppose $V = U \oplus W$.*

1. *Prove that the restriction $\pi|_W : W \to V/U$ is an isomorphism.*
2. *Prove that every $\varphi \in V^*$ can be written uniquely as $\varphi = \varphi_U + \varphi_W$, where φ_U vanishes on W and φ_W vanishes on U.*
3. *Prove that $V^* = U^\circ \oplus W^\circ$.*
4. *Show that $U^\circ \cong W^*$ and $W^\circ \cong U^*$.*